

ELEC-1107 – Project Expectation, Clarification and Rubric

(Project: Powered Speaker and Mobile Device Charging Station)

The project will be a phased approach to applying into practice many of the concepts studied in this electronic devices course. In order to study some “real life” circuit applications, the students will perform design validation and component selection calculations for the project designs. The project involves building a 5 VDC, 3 Ampere USB (Universal Serial Bus) power supply for charging mobile devices. It will also incorporate a powered audio amplifier with a 1/8” audio jack input with a built in speaker. Finally, it will also include some basic safety and monitoring circuits.

The project will be split into 3 sections with a total weight of 30% of the final course grade and will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines. The project will be graded in week 14 with the following grading breakdown:

Section 1 of the project will be the finished product and will be worth 1/3 of the project grade (10% of your final grade). The finished product will be graded on functionality, layout, fabrication techniques and professionalism as outlined in the rubric. Since the students have successfully completed a skills (fabrication and soldering) course, a portion of the fabrication grade will involve soldering all the components on a perforated prototype board.

Section 2 of the project will be an evaluation of the demonstration of the performance characteristic measurements of the project and will be worth 1/3 of the project grade (10% of your final grade). For example, some of the performance characteristic measurements for the power supply portion might include line and load regulation, ripple and ripple factor, etc. The demonstration of the performance characteristic measurements will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

The third and final portion (Section 3) of the project will be in the written lab report and is worth the remaining 1/3 of the grade (10% of your final grade). The lab report will be written with the guidelines as described below and graded against the rubric below. The written lab report portion will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

Section 3 – Formal Report Guidelines

The Beginning of the Paper

This part of the paper generally deals with the question why (was the experiment necessary) by introducing the paper, the background material and the need for the experiment to the reader. The written portion of the project must have all the following content.

Page 1: The Cover Page

The cover or title page should clearly display the name of the experiment, the program code, the course code, the instructors name, your name, the creation date and the submission date.

Page 2: Table of Contents

The table of contents should clearly list all the sections of the report (like the list of figures and tables, the table of contents, the abstract, the introduction/background, etc.) and their pages.

Page 3: Lists of Figures and Tables

The lists of figures and tables should be outlined in two separate lists; one for the figures and the second for tables. They should be listed in order, with a description and the pages they are found on.

Page 4: The Abstract (Summary)

The abstract is an overview of the project. It is usually only a paragraph or two that clearly and concisely presents an overview of the report including what was done, key results or observations and conclusion. The abstract should not be written in point form but in complete sentences.

Page 5 – 8: The Introduction and/or Background Data

In this section, the reader is provided with some historical information. Generally, this section provides the reader with an appreciation as to why you performed the experiment, what was expected as the outcome and an understanding of your interpretation of the results and conclusion. To be effective, this section should include the objective or purpose of the experiment, relevant background information, an indication of application or importance of the work and a brief preview of what is to come. This is also an appropriate place to layout your predictive design calculations.

The Main Body of the Paper

This part of the paper generally answers the question how (was the experiment done and what was learned). This is done by describing the purpose for the experiment, stating your hypothesis, listing the equipment used, describing the procedure in detail, stating the observations, initiating a discussion and providing a detailed conclusion. Each of the sections should be identified by a clearly displayed heading in a logical, systematic manner.

Page 9: Objective

The objective of a paper is to inform the reader what the purpose of the project is. The purpose can be detailed in only a few well constructed sentences in paragraph form. You may paraphrase any information I give you but you cannot copy them directly.

Page 9: Timeline

This section (timeline) along with the next two sections (the equipment and the procedure) form, what is called, “The Method”. The “Method” informs the reader of the general strategy used to collect the data including how the experiment was done, what equipment was used and what you thought the outcome would be prior to performing the work. The timeline details of the timing of the work completed including all major milestones and their dates; such as how much time you spent on the design calculations/simulations, build, troubleshooting, design validation and documentation for this section and when the work was done. The timeline should account for the timing and duration of the project from cradle to grave. At least one timing image, figure, simulation or diagram is required for this section.

Page 10: The Equipment and BoM (Bill of Material)

In this section, a list of the material used in the experiment is provided so that if the reader wished to validate or reproduce your work, they could do so by using the exact same components, equipment and instruments you did. A “Bill of Material” (BoM) table details all the part numbers, descriptions, supplier names, quantities, pricing (in Canadian funds) and shipping details (costs, lead time, etc.). This is done to set assist in setting a project budget. Detail and consistency are key here.

Pages 11 - 13: The Procedure

Remember, in order for any scientific experiment to be considered reputable, the experiment and its results must be repeatable. The key to a repeatable experiment is documentation and so in this section, a detailed account of the procedure for this project completion is listed. The procedural documentation should be subcategorized for each major milestone and expressed systematically. All the calculations performed for the design validation should be included in this step as well. Documenting is not limited strictly to text. Images, figures, sketches, schematic and/or wiring diagrams may accompany your detailed text

based procedural instruction. At least one image, figure, simulation, sketch, schematic or wiring diagram is required for this section.

Results and Discussion

Page 14 - 16: Observations and Analysis

In this section, you will present all relevant nominal (calculated) and experimental information both quantitative and qualitative. Your quantitative results may be expressed graphically in a graph or table and your qualitative observations may be expressed textually with descriptive statements. What form of data representation is most appropriate will typically depend on the nature of the data and the quantity of data points. Many times you will be looking for trends or patterns or identifying the impact of one variable on a (predefined) system. In those cases, expressing your results with a graph is usually appropriate and adequate. However, by adding descriptive statements and calculations to better describe your graphical results may help you find the trends and patterns you were initially looking for and more easily highlight them to the reader. Something to keep in mind when collecting your data and documenting the performance characteristics of the project is asking yourself “Does what I am seeing make sense? Are these results what I would expect? Does what I am observing or what I calculated concur with the predictions made in my preparations or the calculations I performed?” If your observations do not make sense; stop and try to determine why before continuing. Try to determine if the discrepancy you are observing is a result of a miscalculation or an error in the experiment including potential miscalculations or misinterpretations of experimental data.

Page 17: Conclusion and Recommendations (Lessons Learned)

A well constructed conclusion presents your interpretation of your results to the reader. Any conclusion you draw should be clearly and concisely stated and supported with evidence from your observations and calculations. Your conclusion should cite specific results, observations and calculations from the experiment and tie them to the purpose and pre-lab data. Determine and summarize potential reasons for any discrepancies between your measured results and the results you predicted in the preparation. Finally, your conclusion should indicate potential sources of error and recommend any practical way of improving the quality of experiment and data. When drawing your conclusion, try to be as specific and quantitative as possible. Avoid using catch-all phrases such as “human error” or qualitative statements like “the results were quite close”. Be more specific, list the errors or calculate just how “close” your results were. Be sure to comment on “outlying” data points measured during the experiment.

The Final Sections of the Paper

This part of the paper generally answers the question what (information resources are there for the reader to refer to).

Graphs, Tables and Figure/Images

It is important to represent your data as accurately and appropriately as possible and so be sure and determine what type of graph to use. Also, determine whether your data will be best represented by a linear, semi-log or log-log plot. Graphs and tables should be clear, logical and fully labeled so that the reader can interpret your results without having to referring to the text for analysis. Hence, you will have to choose figure captions and table titles carefully. Captions, titles and legends should be descriptive and the axis scales should be appropriate so as to maximize the graph. The data points should be drawn using standard data symbols and connected using the “curve of best fit” plotting technique. Be sure to include plot titles and labels, coordinate labels where appropriate and scale/axis units. Figures, tables and images should be displayed in chronological order and labeled sequentially based on section. The text should reference all graphs, tables, figures and images by number rather than by title. It is okay to use colour however, your plots, sketches, figures and images, etc. should be legible and easy to read and follow in black and white. This means that you shouldn't use colour as your only distinguisher of data.

Page 18: Reference

All references used in your paper must be cited so that the reader can follow up and research the information on their own. In our labs, I will supply the information references in the lab opening section titled “Textbook References”.

The Overall Look and Feel of the Paper

This report (in its entirety), should only be a dozen pages or so in length. The report should be written in a formal style and tone and must be edited for proper spelling, grammar, punctuation, coherence and technical accuracy.

The final copy of the report must be a professionally presented document. It must be word processed; single spaced, and printed on one side of the page only (8.5 x 11). The finished report must be professional in appearance, stapled (top left corner) and presented in sequence. The margins (top, bottom, left and right margins) must be one inch. Keep all graphics and formatting within margins. The report must include sequential page numbering. The title page is not numbered and you should use Roman Numerals for all pages up to and including the abstract.

The report body text must be in 12-point Calibri font. All titles and headings must be in the font. The report title will be in 16-point font. Section headings will vary from 16- to 12-point font size, reflecting hierarchy levels. All text should be justified aligned, except for the title page and some visuals or notes which may be centered on the page. Some elements may be indented, tabbed, right-aligned, or positioned within the text.

The report must be effectively paragraphed. Identify clearly through headers what the subject is being discussed in the paragraph(s) following. Do not hide information inside headers that have no relation to the header. Hierarchy must be clear and consistently formatted. All bulleted and enumerated lists must be introduced using lead-in text.

At least three visual aids (a BoM table, a schematic diagram and oscilloscope data) are required. Visuals must be complete, pertinent to the subject being discussed and must be well integrated into the report. Identify, describe and/or interpret each visual aid. All visual aids should be properly cropped, usefully sized, and cleanly presented. All graphics must be labeled with figure number and descriptive title. Finally, indicate borrowed source for graphics.

All sources used must be properly integrated within the report and documented in a reference page at the end of the report using APA or MLA format. Undocumented reports are incomplete and will not be accepted. A zero grade will be given and an academic penalty may apply.

Please add the following declaration to the top of your Reference page:

I declare that I have performed all the calculations, prepared all the schematic diagrams, created all the simulations, took all the photos, performed all the measurements, recorded all the data, and prepared all aspects of this technical report alone and with no collaboration from anyone or anything (including the internet). If intellectual property from other authors was used (such as datasheet information and/or images, etc.), it is documented and referenced as such regardless if it was taken word for word or paraphrased.

I also declare that I have fully read and understood the project expectations and rubrics and have received clarification on anything I was unsure or unclear of. This includes, but is not limited to, what resources I may use to assist in any section of the project.

Printed Name

Signature

Date

Be sure to sign and date this document or no portion of your project will be graded.